1) Two wings have AR = 8 and use the same symmetric airfoil w/ $a_0 = 2\pi$/rad

- Wing E has an elliptical planform
- Wing R is rectangular. Its lifting line solution has nonzero A_1, A_3, A_5 w/ $\frac{A_3}{A_1} = 0.12$, $\frac{A_5}{A_1} = 0.03$ & $\tau = 0.18$

a) For wing E, find its lift slope α and C_L at $\alpha = 4^\circ$

b) For wing E, find α_i & α_{eff}

c) For wing E, Find A_1

d) For wing R, find e_w, lift slope α & C_L at $\alpha = 4^\circ$

e) Both wings must fly at $C_L = 0.5$. For each wing, find the required α and $C_{D,i}$. Which way has less induced drag & by what percent?

a) $$a = \frac{a_0}{1 + \frac{a_0}{\pi AR}(1+\tau)} = \frac{2\pi}{1 + \frac{2\pi}{8\pi}} = \frac{2\pi}{1.25} = 5.03/\text{rad}$$

$$C_L = a\alpha = 5.03 \cdot 0.0698 = 0.351$$

b) $$\alpha_i = \frac{C_L}{\pi AR} = 0.8^\circ$$

$$\alpha_{eff} = \alpha - \alpha_i = 4 - 0.8 = 3.2^\circ$$

c) $$C_L = A_1 \pi AR \rightarrow A_1 = \frac{C_L}{\pi AR} = \frac{0.351}{25.13} = 0.0140$$

d) $e_w = \dfrac{1}{1 + \sum_{2}^{N} n \left(\frac{A_n}{A_1}\right)^2} = \dfrac{1}{1 + 3(0.12)^3 + 5(0.03)^2}$

$= \boxed{0.955}$

$$a = \frac{a_0}{1 + \frac{a_0}{\pi AR}(1+\tau)} = \frac{2\pi}{1 + \frac{2\pi}{8\pi}(1.18)} = 4.85 \text{ /rad}$$

$$C_L = a\alpha = 4.85\,(0.0698) = 0.339$$

e) Elliptical wing:

$C_L = a\alpha$

$0.5 = 5.03\alpha$

$\alpha = 5.7^\circ$

$$C_{D_i} = \frac{C_L^2}{\pi e_w AR} = \frac{0.5^2}{\pi(1)(8)} = 0.00995$$

Rectangular: $\alpha = 5.9^\circ$

$$C_{D_i} = \frac{0.5^2}{\pi(0.955)(8)} = 0.01042$$

2) An airplane has weight $W = 50{,}000\ N$, $S = 30\ m^2$, $AR = 9$, $C_{D,0} = 0.025$. It flies at sea level w/ $\rho = 1.225\ kg/m^3$.

In level flight $L = W$

a) Find the Oswald efficiency factor

b) Find $\left(\frac{L}{D}\right)_{max}$ and the C_L where it occurs. Show that at this C_L, the drag due to lift equals $C_{D,0}$

c) Find the flight speed and the drag force at $\left(\frac{L}{D}\right)_{max}$

d) At 90 m/s, Find C_L, C_D, L/D & the drag force. What fraction of C_D is $C_{D,0}$

a) $e_o = 1.78\left(1 - 0.0045\ AR^{0.68}\right) - 0.64$

$e_o = 1.78\left(1 - 0.0045 \times 9^{0.68}\right) - 0.64 = 0.783$

b) $C_L = \sqrt{\pi\, e_o\, AR\, C_{D,0}}$

$= \sqrt{\pi(0.783)(9)(0.25)} = 0.744$

$\left(\frac{L}{D}\right)_{max} = \frac{\sqrt{\pi\, e_o\, AR\, C_{D,0}}}{2\, C_{D,0}} = \frac{0.744}{2(0.025)} = 14.9$

$C_{D_i} = \frac{C_L^2}{\pi e_o AR} = \frac{0.554}{22.14} = 0.025 = C_{D,0}$ ✓

c) $L = W$

$$W = \tfrac{1}{2}\rho V^2 S C_L \rightarrow V = \sqrt{\frac{2W}{\rho S C_L}} = \boxed{60.5 \text{ m/s}}$$

$$\frac{L}{D} = \left(\frac{L}{D}\right)_{max} \rightarrow D = \frac{W}{(L/D)_{max}} = \frac{50\,000}{14.9}$$

$$\boxed{D = 3360 \text{ N}}$$

d) $C_L = \dfrac{2W}{\rho V^2 S} = \dfrac{100{,}000}{1.225 \cdot 90^2 \cdot 30} = \boxed{0.336}$

$$C_D = C_{D_0} + \frac{C_L^2}{\pi e_0 AR} = 0.025 + \frac{0.336^2}{\pi(0.783)9}$$

$$= 0.0301$$

$$\frac{L}{D} = \frac{0.336}{0.0301} = 11.2 \qquad D = \frac{W}{(L/D)} = \frac{50{,}000}{11.2}$$

448 N

$$C_{D,0} = \frac{0.025}{0.031} = 83\%$$

3) An airplane flies at $V_\infty = 200$ m/s when $T_\infty = 250$ K & $P_\infty = 50{,}000$ Pa. Take $\gamma = 1.4$ & $R = 287$ J/kg·K

a) Find ρ_∞ & a

b) Find the compressibility τ

c) Find M_∞. What flow regime is this?

d) Find stagnation pressure

e) A low speed wind tunnel test gave $C_{p,0} = -0.45$ at one point in the wing. Can Prandtl-Glauert be used here? If so, find C_p & the actual pressure P at this point.

a) $\rho_\infty = \frac{P}{RT} = \frac{50{,}000}{287 \times 250} = 0.697$ kg/m^3

$a = \sqrt{\gamma R T} = \sqrt{1.4(287)(250)} = 316.9$ m/s

b) $\tau = \frac{1}{\rho a^2} = \frac{1}{0.697\,(316.9)^2} = 1.43 \times 10^{-5}\ \frac{1}{Pa}$

c) $M_\infty = \frac{200}{316.9} = 0.631 \rightarrow$ Compressible Subsonic

d) $\frac{P_0}{P_\infty} = \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}} = \left(1 + \frac{1.4-1}{2} 0.631^2\right)^{\frac{1.4}{1.4-1}}$

$= 1.308\,(P_\infty) = \boxed{65{,}400}$

e) Yes, $M_\infty < 0.7$

$$C_p = \frac{C_{p_0}}{\sqrt{1 - M_\infty^2}} \qquad C_p = \frac{-0.45}{\sqrt{1 - 0.631^2}} = -0.580$$

$$C_p = \frac{2}{\gamma M_\infty^2}\left(\frac{P}{P_\infty} - 1\right)$$

$$P = P_\infty\left(1 + \frac{\gamma M_\infty^2}{2} C_p\right)$$

$$= \boxed{41{,}800 \text{ Pa}}$$